

- Q.17 Define center of gravity. How is it different from centroid?
- Q.18 Why is paper considered an important part of modern life? State any 2 uses of paper.
- Q.19 Briefly explain the terms-sulphidity, Causticity
- Q.20 Write a short note on middle lamella of fiber.
- Q.21 Briefly discuss the bleaching stage of paper making. What is its importance?
- Q.22 Define unit process. Give 2 examples each of Oxidation, Hydrogenation.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Derive the expression for the resultant of two forces acting at a point using the parallelogram law of forces.
- Q.24 Why is bamboo considered as the most important raw material for paper making? Explain its physical and chemical characteristics.
- Q.25 Describe the manufacturing process of cement with the help of a block diagram.

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1st Year / Chemical Engg. (P&P)

Subject : Introduction to Chemical Engg and Paper Technology

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 For paper industry, the large share of raw material comes from
- a) Agricultural waste
 - b) Bamboo and wood
 - c) recycled paper
 - d) Petroleum by product
- Q.2 Which of the following is correct
- a) paper=thin sheets
 - b) pulp= fibrous raw material for paper making
 - c) both a and b
 - d) none of the above

Q.3 The paper industry in India is considered important because

- a) it contributes to literacy and education
- b) it supports packaging and industrial sector
- c) it generates employment opportunities
- d) all of the above

Q.4 Bamboo is classified as

- a) non wood fibrous raw material
- b) hard wood
- c) soft wood
- d) mineral additive

Q.5 Main chemical component of all paper making fibers is

- a) cellulose b) lignin
- c) hemicellulose d) silica

Q.6 Which of the following is not a unit operation

- a) distillation b) filtration
- c) polymerisation d) Crystallisation

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SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 1 K Cal = _____ joules.

Q.8 Convert 100° C into Kelvin temperature.

Q.9 Green liquor in the recovery process mainly contains _____

Q.10 Define cooking cycle in pulping.

Q.11 What is the role of lignin in the plants?

Q.12 What are the main sugar units present in hemicellulose?

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Define paper technology. Explain its importance in modern society.

Q.14 Discuss the scope of pulp and paper technology in terms of employment opportunities.

Q.15 List four important factors that influence the location of a paper industry.

Q.16 Define thermal conductivity. Give 2 examples each of material with high thermal conductivity and low thermal conductivity.

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